Real-Time Smart Energy Management System

Optimizing source selection between grid and renewable/battery power using real-time sensing and time-of-use pricing.

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Problem Definition & Objectives

Static transfer switches protect uptime, but they do not optimize energy cost or source usage.

The problem

Peak-rate periods make grid energy more expensive.

Conventional transfer switches are mainly reactive: they fail over when power is lost, but they do not evaluate cost, load demand, or battery condition.

A practical EMS needs to choose the source that is safe, available, and cost-aware.

Project objectives

1 Monitor

Read grid, battery, and load conditions with near real-time telemetry.

2 Decide

Apply time-of-use pricing and battery health thresholds to select a source.

3 Switch

Drive relays safely with interlocks, dwell timing, and break-before-make transfer.

Success criteria for the demo

- Dashboard updates within ~1 second
- Source transfer time under 250 ms target
- Automatic battery use during expensive peak-rate periods when battery is healthy
- Manual override available without defeating protection logic

Design intent: reduce peak-rate grid usage while maintaining reliable load operation.

System Architecture

TECHNICAL APPROACH

Four modules: power sources, primary processing, load output, and remote host/dashboard.

Power sources

12 V grid supply, solar panel, and 12–14.2 V SLA battery through charge control.

Primary processing

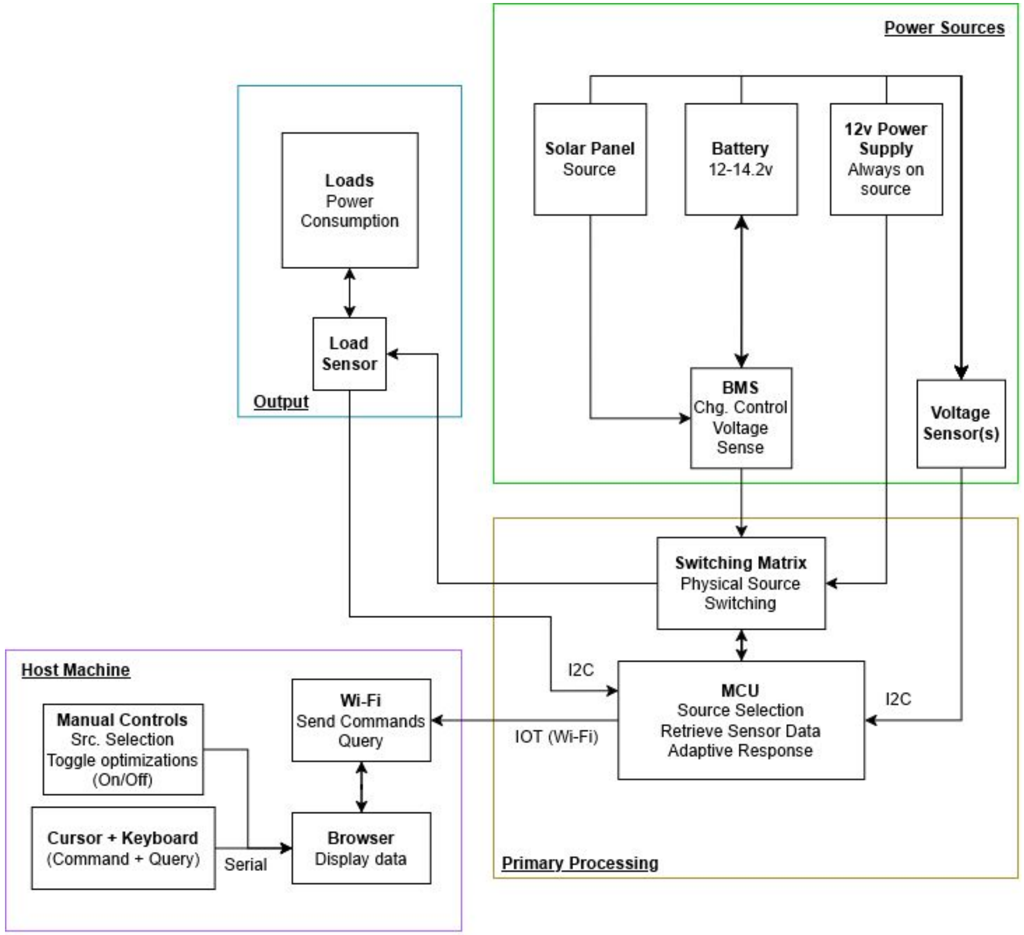
ESP32 reads sensors, evaluates source logic, and commands the relay switching matrix.

Output + feedback

Load receives power from the selected source; load sensor feeds demand data back to the controller.

Host interface

Arduino IoT dashboard displays system status and supports manual source control.

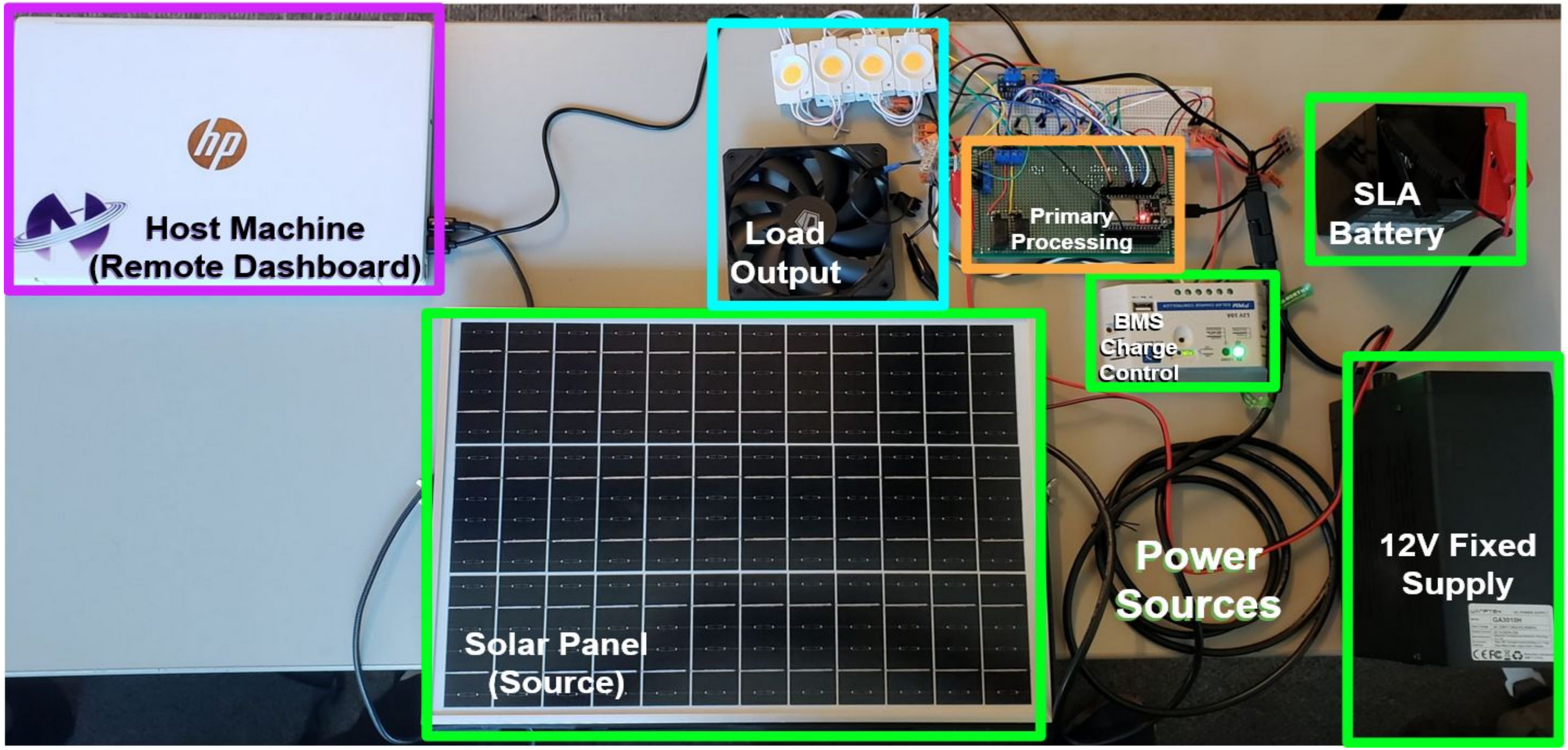


Energy Management System
Dynamic Source Optimization

System Block Diagram

Physical System Layout

Full system — solar panel, SLA battery, BMS charge controller, ESP32, relay matrix, load output, and remote host machine



Decision Logic: Cost-Aware, Battery-Aware Switching

TECHNICAL APPROACH

The EMS only uses the battery when it is valuable and safe to do so.

1 Sense

Load current
Battery voltage/SOC
Current price band

2 Evaluate

Peak-rate?
Battery healthy?
Manual override?

3 Latch

Confirm stable decision for 5 simulated minutes

4 Switch

Apply interlocked relay state

Off-peak / cheap period

Use grid by default

Peak / expensive period

Use battery if healthy

Weak battery / unsafe

Fall back to grid

Battery hysteresis

Enter battery mode only when SOC ≥ 35% and V ≥ 12.20 V.
Remain on battery while SOC ≥ 25% and V ≥ 12.00 V.

Riverside summer TOU model

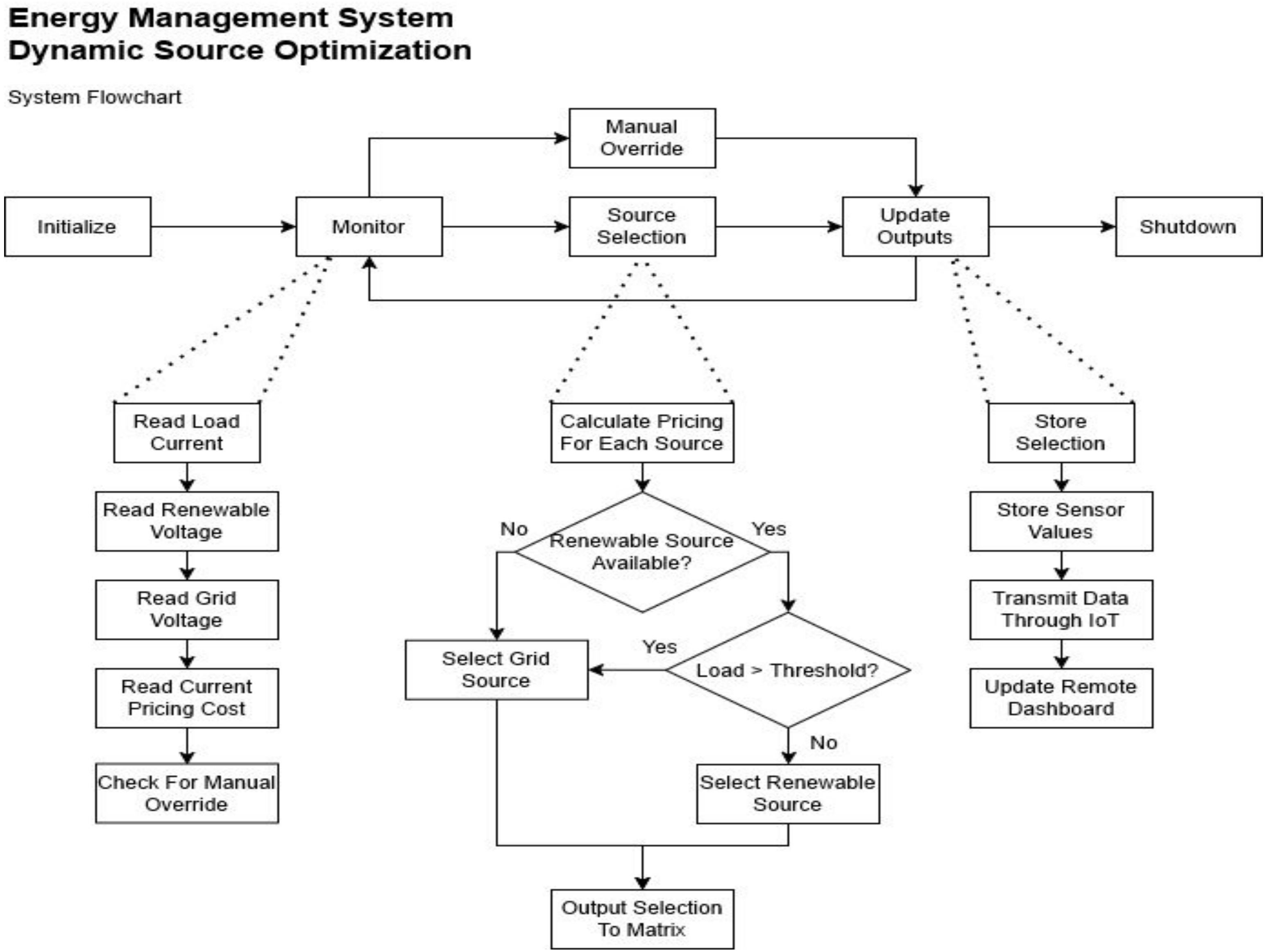
Peak: 4 PM–8 PM at $0.55/kWh.
Off-peak: $0.22/kWh.

Anti-oscillation

A changed decision must remain stable before the relays are allowed to transfer.

Decision Logic: System Flowchart

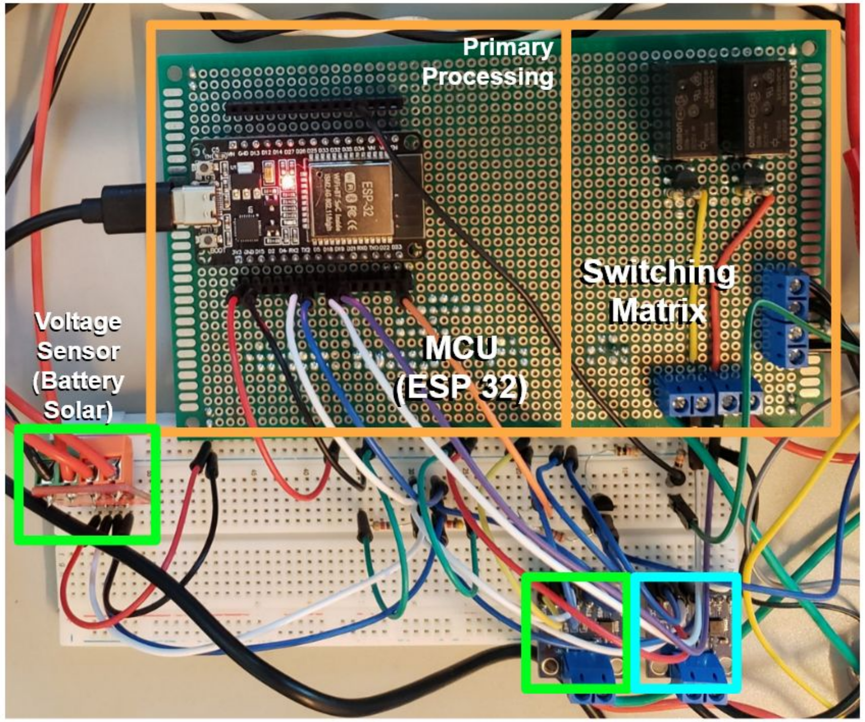
Initialize → Monitor → Source Selection → Update Outputs, with manual override available at any point



Technical Execution: Hardware, Sensing, and Protection

TECHNICAL EXECUTION

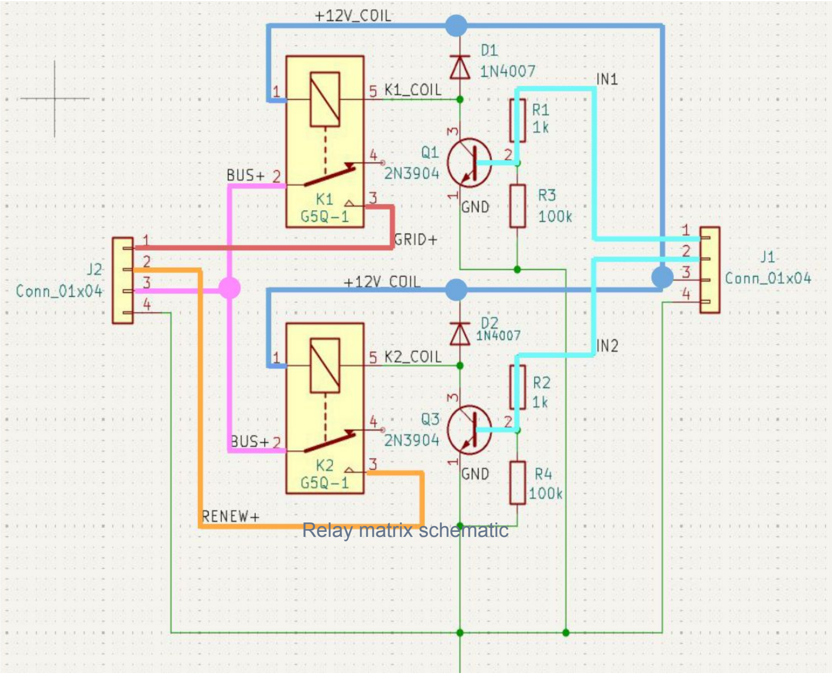
The final build prioritizes stable sensing and safe relay transfer.



Voltage Sensor (Grid)

Load Sensor (Output)

Primary processing module: ESP32 + sensors + switching matrix



Relay matrix schematic

Sensing

INA226 modules monitor source/load voltage and current over I²C.

Switching

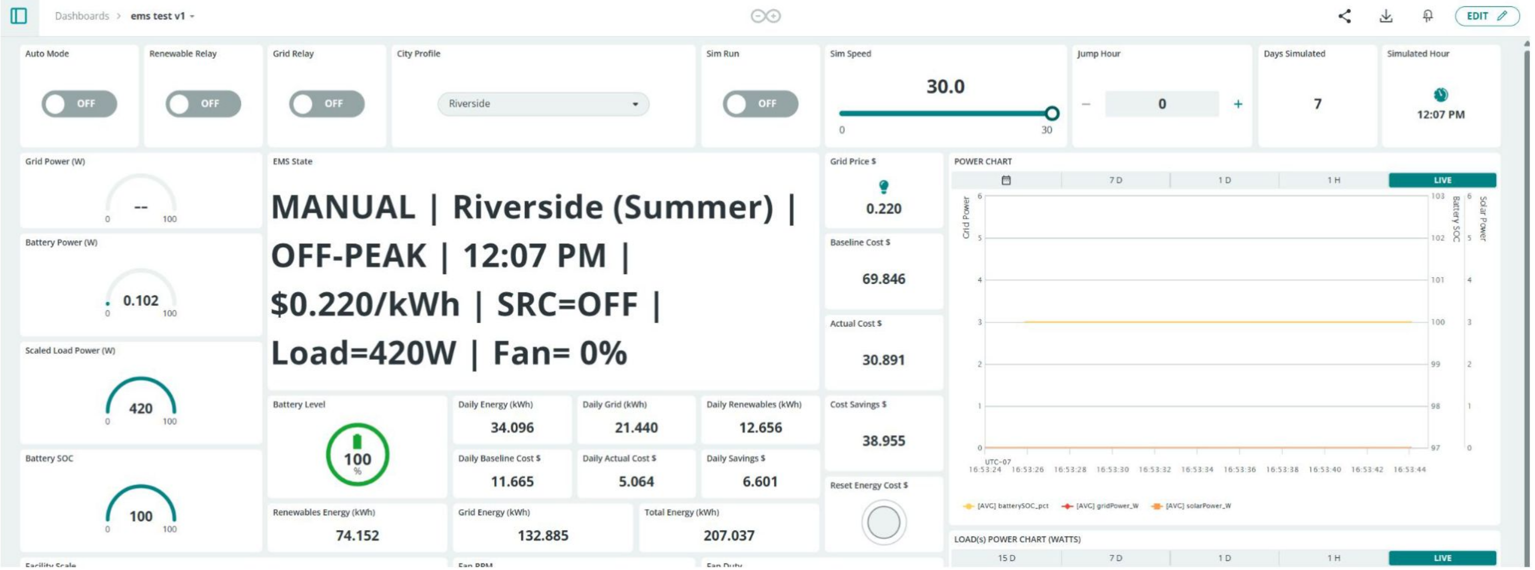
12 V relays select either grid or battery/renewable path.

Protection logic: relay interlock • 150 ms break-before-make • 600 ms automatic dwell • 2500 ms manual dwell

Results & Validation

RESULTS

The final system met all core demonstration targets — including a ~40% projected cost reduction over 30 simulated days.



Arduino IoT Cloud dashboard used for live monitoring and manual control

800–1000 ms

Telemetry update window; target was within about 1 second.

<150 ms

Measured source transfer time; target was under 250 ms.

600 ms

Automatic dwell time used to stabilize relay behavior.

5.8 A

Overcurrent warning/protection triggered near threshold.

~$176

Estimated build cost.

~40%

Projected cost reduction over 30 simulated days vs. grid-only.

Validation summary

Verified I²C sensor presence, near-real-time dashboard telemetry, manual and automatic relay switching, peak-rate battery selection, and fail-safe fallback behavior.

Conclusion

A functional EMS prototype that senses, decides, switches, and reports in real time.

What we achieved

Autonomous grid/battery source selection using TOU pricing, battery thresholds, and live load sensing. All timing and protection targets met. ~40% projected cost reduction over 30 simulated days vs. always using the grid.

Why it matters

Shows how small-scale embedded control can reduce peak-rate grid usage while preserving load reliability.

Bottom line: the project successfully demonstrates safe, real-time, cost-aware source management.

Demo link: https://youtu.be/YzSE4Eos70A